

School: SET		Batch: 2024-2028
Program: B Tech.		Current Academic Year: 2026-27
Branch: CSE		Semester: V
1	Course Code	CSE021
2	Course Title	Introduction to Cloud Computing
3	Credits	3
4	Contact Hours (L-T-P-SL)	3-0-0-3
	Course Status	Program Elective
5	Course Objective	This introductory course on Cloud computing will teach both the fundamental concepts of how and why Cloud systems works, as well as Cloud technologies that manifest these concepts.
6	Course Outcomes	At the end of the course, students will have achieved the following learning objectives. CO1. Define the basics of cloud and recall the computer Science concepts which are helpful in understanding on demand service architecture. CO2. Classify and describe the architecture and taxonomy of parallel and distributed computing, including shared and distributed memory, and data and task parallel computing. CO3. Apply the PAAS and SAAS to manage the workflow and use of cloud in scientific application. CO4. Categorize and Characterize between Infrastructure services, deployment models, and governance in cloud computing. Examine the design of task and data parallel distributed algorithms for Clouds and use them to construct Cloud applications. CO5. Evaluate the importance of cloud using monitoring and management of services for performance improvement of HPC and to follow the Governance and Compliances. CO6. Elaborate the design concept and formulate to build the solution using cloud service providers as AWS, MS Azure, Google Cloud. Demonstrate the use of Map-Reduce, Vertex-Centric and Continuous Dataflow programming models.
7	Course Description	This course is an introductory course for cloud computing concepts and helps in understanding the core functionalities, algorithms, models and workflows in cloud environment. In this course Students will get demonstrations of real-time cloud services for better exposure and research understanding.

8	Teaching and Learning Scheme (in Hours per Semester)	Classroom Instructions (CI)				Lab Instructions (LI)	Teamwork (TW) and Self Learning (SL) (TW+SL)	Total number of Hours	Total Credits TH/30
		L		T		P	SL	TH	
		45		0		0	45	90	3
9	Outline syllabus								CO Mapping
	Unit 1	FOUNDATIONS							
	A	Introduction to compute Types of Computing, Grid computing, distributed computing, Client-server computing, Three Tier Architecture, use of Sockets and Remote Procedure Call, working of RMI and CORBA, Web services, Web Sockets, Message Queues and Message Brokers.							CO1
	B	Introduction to Cloud Computing Cloud Computing definition, Roots of Cloud Computing, Layers and Types of Clouds, Desired Features of a Cloud, Cloud Infrastructure Management, Infrastructure as a Service Providers, Platform as a Service Providers, Challenges and Risks							CO1
	C	Migrating and Integrating into Cloud Broad Approaches to Migrating into the Cloud, The Seven-Step Model of Migration into a Cloud, Enriching the ‘Integration as a Service’ Paradigm for the Cloud Era, Evolution and Challenges of SaaS Paradigm, Integration Scenarios, The Integration Methodologies							CO1
	Unit 2	ENTERPRISE CLOUD COMPUTING AND IAAS							
	A	The Enterprise Cloud Computing Paradigm Issues for Enterprise Applications on the Cloud, Transition Challenges, Enterprise Cloud Technology and Market Evolution, Business Drivers Toward a Marketplace for Enterprise Cloud Computing, The Cloud Supply Chain							CO1, CO2
	B	Virtual Machines Provisioning and Migration Services Introduction to Virtual Machines, The Anatomy of Cloud Infrastructures, VM Provisioning and Manageability, Virtual Machine Migration Services, Management of Virtual Machines for Cloud Infrastructures, Distributed Management of Virtual Infrastructures, Scheduling Techniques							CO1, CO2
	C	Enhancing Cloud Computing Environments Using a Cluster as a Service							CO1, CO2

		Introduction and Related Work, RVWS Design, Cluster as a Service: The Logical Design, Secure Distributed Data Storage in Cloud Computing, Cloud Storage, Technologies for Data Security in Cloud Computing	
	Unit 3	PLATFORM AND SOFTWARE AS A SERVICE	
	A	Aneka and Comet Cloud Aneka—Integration of Private and Public Clouds, Technologies and Tools for Cloud Computing, Aneka Cloud Platform, Comet Cloud: An Autonomic Cloud Engine, Introduction of Comet Cloud (Architecture, Autonomic Behavior, Applications overview)	CO1, CO3
	B	Business Solutions and Workflow Cloud-Based Solutions for Business Applications (Introduction of Enterprises Demand and Cloud Computing, Dynamic ICT Services), Workflow Engine for Clouds, Workflow Management Systems, Architecture of Workflow Management Systems	CO1, CO3, CO6
	C	Scientific Applications and Map Reduce Model Scientific Application for Cloud Environments, Classification of Scientific Applications and Services in the Cloud, SAGA-based Scientific Applications, Map Reduce Programming Model, Map Reduce Impacts and Research Directions	CO1, CO3, CO6
	Unit 4	MONITORING, MANAGEMENT & GOVERNANCE	
	A	SLA Management in Cloud Computing Introduction of typical Use Cases, Model for Federated Cloud Computing, Security Considerations, SLA Management in Cloud Computing: A Service Provider's Perspective, Types of SLA, Life Cycle of SLA, Automated Policy-based Management	CO1, CO4
	B	Performance Predictions for HPC on Clouds Introduction and Background of Grid and Cloud, HPC in the Cloud: Performance-related Issues, Game Hosting on Cloud Resources, Building Content Delivery Networks Using Clouds, Resource Cloud Mashups	CO1, CO4
	C	Security and Governance Basic Concept of Organizational Readiness, Drivers for Changes: Common Change Management Models, Security and Risk in the Cloud, Cloud Computing and Identity, Content Level Security—Pros and Cons, Legal Issues in Cloud Computing(PCI DSS), Data Privacy and Security Issues	CO1, CO4

	Unit 5	AWS, MS AZURE AND GOOGLE CLOUD																							
	A	AWS Services:EC2, IAM, S3, Lambda, EBS, CDN, Cloud Watch,			CO1, CO5, CO6																				
	B	MS Azure Services: Azure VM , SQL Server on Virtual Machines, Azure SQL Database, Azure Active Directory, Azure Backup			CO1, CO5, CO6																				
	C	Google Cloud: Compute Engine, Migrate for Compute Engine, Cloud Functions, Gsuite Admin, Cloud Lab Balancing, Cloud Storage			CO1, CO5,CO6																				
10	List of activities for Self-Learning (SL: 45h)	<table><tr><td>S. No.</td><td>Name of the Activity</td><td>No of Hours</td><td colspan="2">Evaluation Criteria</td></tr><tr><td>1</td><td>Assignment writing on estimation, scheduling, earned value, and risk analysis</td><td>5 assignments of 3h each. Total = 15h</td><td colspan="2">Based on the assignment submitted.</td></tr><tr><td>2</td><td>Case Study Based Learning</td><td>Case Study Selection & Research =5h Analysis & Report Writing = 7h Presentation Preparation = 3 h; Total = 15h</td><td colspan="2">Correct identification and use of relevant cloud based models. Use of standard references, textbooks, NPTEL, IEEE/Springer papers</td></tr><tr><td>3</td><td>Technical Video based learning related to software project management</td><td>Duration of video = 5h; Report preparation & Presentation = 10h; Total = 15h</td><td colspan="2">Report/presentation based on the video learning outcomes.</td></tr></table>				S. No.	Name of the Activity	No of Hours	Evaluation Criteria		1	Assignment writing on estimation, scheduling, earned value, and risk analysis	5 assignments of 3h each. Total = 15h	Based on the assignment submitted.		2	Case Study Based Learning	Case Study Selection & Research =5h Analysis & Report Writing = 7h Presentation Preparation = 3 h; Total = 15h	Correct identification and use of relevant cloud based models. Use of standard references, textbooks, NPTEL, IEEE/Springer papers		3	Technical Video based learning related to software project management	Duration of video = 5h; Report preparation & Presentation = 10h; Total = 15h	Report/presentation based on the video learning outcomes.	
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11	Mode of examination	Theory																							
12	Weightage Distribution	CA	MSE	ESE																					
		25%	15%	60%																					
13	Text book/s*	1. CLOUD COMPUTING Principles and Paradigms, Edited by Raj Kumar Buyya, Jam 2. Cloud Computing: A Practical Approach, Anthony T. Velte, Toby J. Velte, Robert Elsenpeter																							
	Other References																								

CO and PO Mapping

S. No.	Course Outcome	Program Outcomes (PO) & Program Specific Outcomes (PSO)
CO1:	Define the basics of cloud and recall the computer Science concepts which are helpful in understanding on demand service architecture.	PO1, PO2, PSO1
CO2:	Classify and describe the architecture and taxonomy of parallel and distributed computing, including shared and distributed memory, and data and task parallel computing.	PO1, PO2, PO3, PO5, PSO1, PSO2
CO3:	Apply the PAAS and SAAS to manage the workflow and use of cloud in scientific application.	PO1, PO2, PO3, PO5, PSO1, PSO2
CO4:	Categorize and Characterize between Infrastructure services, deployment models, and governance in cloud computing. Examine the design of task and data parallel distributed algorithms for Clouds and use them to construct Cloud applications.	PO1, PO2, PO3, PO4, PO5, PSO1, PSO2
CO5:	Evaluate the importance of cloud using monitoring and management of services for performance improvement of HPC and to follow the Governance and Compliances.	PO1, PO2, PO3, PO4, PO5, PSO1, PSO2
CO6:	Elaborate the design concept and formulate to build the solution using cloud service providers as AWS, MS Azure, Google Cloud. Demonstrate the use of Map-Reduce, Vertex-Centric and Continuous Dataflow programming models.	PO1, PO2, PO3, PO4, PO5, PSO1, PSO2, PSO3

PO and PSO mapping with level of strength for Course Name Introduction to Cloud Computing (Course Code CSE021)

Course Code_ Course Name	COs \ POs/PSOs	PO 1	PO 2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO1 0	PO1 1	PSO 1	PSO 2	PSO 3
Introducti on to Cloud Computin g (Course Code CSE021)	CO1	3	2	–	–	1	–	–	–	–	–	1	2	1	–
	CO2	3	3	2	1	2	–	–	–	–	–	1	3	2	–
	CO3	2	2	3	1	3	–	–	–	1	1	–	3	2	1
	CO4	2	3	2	2	2	–	–	–	–	1	–	3	2	1
	CO5	2	3	2	3	2	1	–	–	–	1	1	3	2	–
	CO6	2	2	3	2	3	–	–	–	1	1	1	3	3	2

Average of non-zeros entry in following table (should be auto calculated).

Course Code	PO 1	PO2	PO 3	PO 4	PO 5	PO 6	PO 7	PO 8	PO 9	PO 10	PO 11	PSO 1	PSO 2	PSO 3
CSE021	2.33	2.5	2.4	1.8	2.17	1	0	0	1	1	1	2.83	2	1.33

Strength of Correlation: 21.36

1. Addressed to Slight (Low=1) extent
2. Addressed to Moderate (Medium=2) extent
3. Addressed to Substantial (High=3) extent

Course-to-SDG Matrix

Sustainable Development Goals (SDGs) are 17 global targets established by the United Nations in 2015 to end poverty, protect the planet, and ensure prosperity and peace for all by 2030. The course Introduction to Cloud Computing equips students with knowledge of cloud architectures, virtualization, distributed computing, cloud service models, and modern cloud platforms such as AWS, Microsoft Azure, and Google Cloud. The course aligns strongly with SDG 4 and SDG 9 by fostering technical education and supporting innovation-driven digital infrastructure development. It also contributes to SDG 8 through skill development in cloud technologies that are highly relevant to current industry demands and employment opportunities. Additionally, cloud-enabled services and scalable digital systems support sustainable urban infrastructure under SDG 11, while governance and compliance concepts modestly contribute toward trustworthy and secure digital environments aligned with SDG 16.

Mapping uses a 3-point scale:

- Addressed to **Substantial (High=3) extent**: Core content directly contributes
- Addressed to **Moderate (Medium=2) extent**: Supports through skills/activities
- Addressed to **Slight (Low=1) extent**: Indirect awareness/benefit

SDG Goal	SDG Description	Mapping Strength	Justification
SDG 4	Quality Education	3	The course provides foundational and advanced knowledge of cloud computing, distributed systems, virtualization, and cloud service models, thereby strengthening quality technical education and digital competencies.
SDG 8	Decent Work and Economic Growth	2	Practical exposure to cloud platforms such as AWS, Microsoft Azure, and Google Cloud enhances employability and prepares students for cloud-centric industry roles and digital transformation initiatives.
SDG 9	Industry, Innovation and Infrastructure	3	The course strongly supports innovation and resilient digital infrastructure through scalable cloud architectures, distributed computing models, HPC, and cloud-based application development.

SDG 11	Sustainable Cities and Communities	2	Cloud-enabled technologies contribute to smart city infrastructure, intelligent resource management, and scalable digital services for sustainable urban development.
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Total Mapped SDGs: 4 out of 17.